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# Mini Project: Wheel Position Control Simulation

## Table of Contents

|                                                    |   |
|----------------------------------------------------|---|
| Motor Model Parameters .....                       | 1 |
| Controller Settings .....                          | 1 |
| Run the Simulink Model .....                       | 1 |
| Plot the Position Response and Motor Voltage ..... | 2 |
| Interpretation of Results .....                    | 3 |

Cailey Cashion EENG 350 - SEED Lab

This script runs a Simulink model of the wheel position controller and plots the desired position, simulated position, and motor voltage command.

The position PI controller uses position error to calculate a desired wheel speed. The inner proportional velocity controller uses speed error to calculate the motor voltage command.

Required file: Simulink\_miniProject.slx The model must be in the current folder or on the MATLAB path.

These results are simulated responses, not experimental measurements.

## Motor Model Parameters

The motor model relates applied voltage to angular velocity:

$$G(s) = K \cdot \sigma / (s + \sigma)$$

```
K = 2.2;  
sigma = 5;
```

## Controller Settings

Kp is the proportional gain of the inner velocity controller. It determines how strongly the voltage command responds to the difference between desired and simulated wheel speed.

Kp is a controller gain; it is different from the motor-model gain K.

The outer position PI gains are configured in the Simulink controller block and are not assigned by this script.

```
Kp = 2.0;
```

## Run the Simulink Model

Run the model using the parameters defined above. Simulation duration and the position step are configured in the model.

The model returns these logged signals as timeseries: out.DesiredPosition - commanded wheel angle in radians  
out.Position - simulated wheel angle in radians  
out.Voltage - simulated motor voltage command in volts

```
out = sim('Simulink_miniProject');
```

*Warning: The file containing block diagram 'Simulink\_miniProject' is shadowed by a file of the same name higher on the MATLAB path. This can cause unexpected behavior. For more information see [matlab:helpview\(\[docroot '/toolbox/simulink/helptargets.map'\], 'shadowing'\)](matlab:helpview([docroot '/toolbox/simulink/helptargets.map'], 'shadowing')) "Avoiding Problems with Shadowed Files" in the Simulink documentation.*

*The file containing the block diagram is:  
C:\Users\cashi\Downloads\Simulink\_miniProject.slx.  
The file higher on the MATLAB path is: C:\Users\cashi\Downloads\Simulink Mini Project\Simulink\_miniProject.slx*

## Plot the Position Response and Motor Voltage

The upper plot compares the desired and simulated wheel positions. It shows how quickly the wheel approaches its target and whether the response overshoots, oscillates, or retains a position error.

The lower plot shows the voltage command used to produce that motion.

```
figure('Name', 'Mini Project: Position Control', 'Color', 'w');

% Position response.
subplot(2,1,1);

plot(out.DesiredPosition.Time, out.DesiredPosition.Data, ...
     '--', 'LineWidth', 1.5);
hold on;

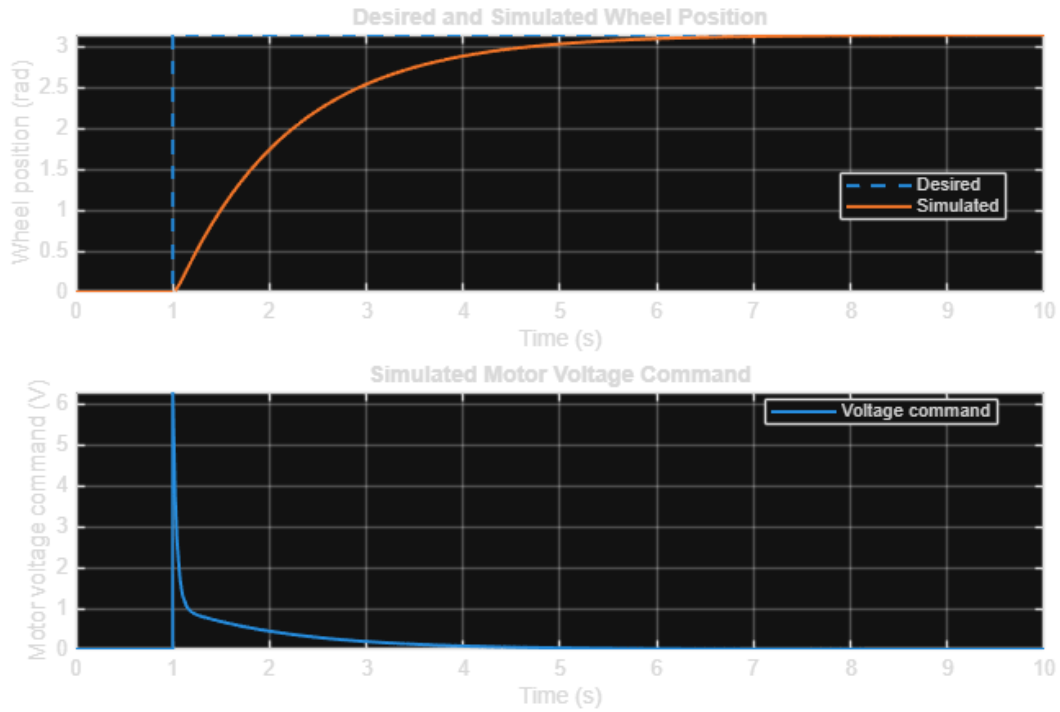
plot(out.Position.Time, out.Position.Data, ...
     'LineWidth', 1.5);

grid on;
xlabel('Time (s)');
ylabel('Wheel position (rad)');
title('Desired and Simulated Wheel Position');
legend('Desired', 'Simulated', 'Location', 'best');

% Motor voltage command.
subplot(2,1,2);

plot(out.Voltage.Time, out.Voltage.Data, ...
     'LineWidth', 1.5);

grid on;
xlabel('Time (s)');
ylabel('Motor voltage command (V)');
title('Simulated Motor Voltage Command');
legend('Voltage command', 'Location', 'best');
```



## Interpretation of Results

For the simulation shown during testing, the desired position changed from zero to approximately  $\pi$  radians at  $t = 1$  second. This shows a half rotation of the wheel.

The simulated position approached the target smoothly, with no overshoot or oscillation in the plotted response.

The voltage command initially rose to around 6 V and then decreased toward zero as the wheel approached the target.

This response supports stable position tracking for the simulated test.

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